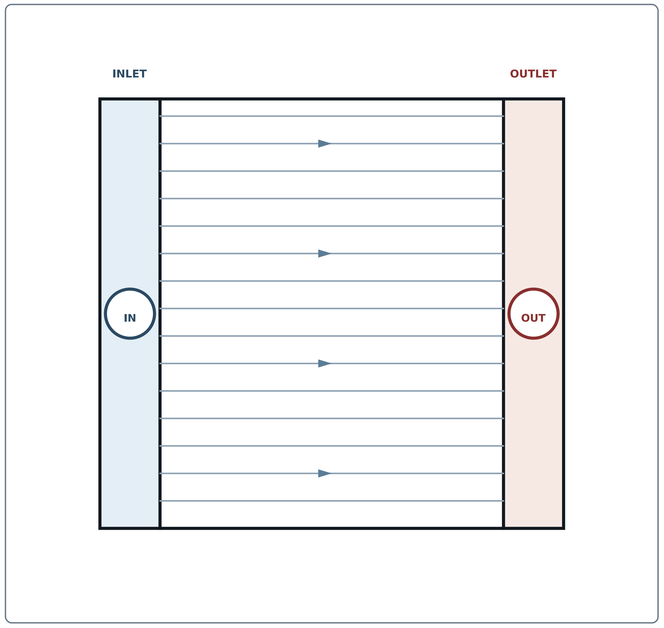
FLOW PATH

SCALE 1.20:1



125 PARALLEL CHANNELS — 0.25 W x 4.0 H x 40 LONG

Single pass, straight. No U-turn, no serpentine:

at 0.25 mm a turn costs more than the whole fin field.

DESIGN POINT — 1.5 L/min, 700 W

PARAMETER	WATER	PG25
velocity in channel	0.20 m/s	**0.20 m/s**
Reynolds number	170	**71**
heat transfer coefficient	7641 W/m2K	**5300 W/m2K**
fin efficiency	0.47	**0.55**
coolant temperature rise	6.8 K	**7.0 K**
pressure drop, plate only	2.5 kPa	**6.1 kPa**

ROBUSTNESS — CONSTANT FLOW

CHANNEL BLOCKAGE	OPEN	VELOCITY	R total	DIE @45 C	dP
all clear	**125**	**0.20 m/s**	**0.0295**	**65.7 C**	**6.1 kPa**
5 % blocked	118	0.21 m/s	0.0299	66.0 C	6.5 kPa
10 % blocked	112	0.22 m/s	0.0303	66.2 C	6.9 kPa
20 % blocked	100	0.25 m/s	0.0313	66.9 C	7.7 kPa

READING THESE NUMBERS

- PG25 is the design fluid. Water is shown only to bound the gain from inhibited water where freeze protection is not required — about 5 % on total resistance.
- A SINGLE blocked channel is not a thermal event. The 1.5 mm copper base carries its ~10 W sideways 0.4 mm to the neighbours for roughly 0.1 K.
- Progressive fouling across many channels is the real risk, and it shows up first as rising pressure drop, not as a rising die temperature.
- So instrument the loop for DIFFERENTIAL PRESSURE, not only temperature, and filter to 50 um at the CDU. Service trigger: dP 20 % above the commissioning baseline at the same flow.

FLOW SENSITIVITY — PG25, 700 W

FLOW	R total	DIE @45 C	RISE	dP
0.75 L/min	0.0373	71.1 C	14.0 K	3.1 kPa
1.00 L/min	0.0333	68.3 C	10.5 K	4.1 kPa
1.50 L/min	**0.0295**	**65.7 C**	**7.0 K**	**6.1 kPa**
2.00 L/min	0.0278	64.4 C	5.2 K	8.2 kPa
3.00 L/min	0.0261	63.3 C	3.5 K	12.3 kPa

Below 1.0 L/min the plate loses ground quickly. Above 2.0 L/min it buys almost nothing — spend the pump head elsewhere in the loop.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK
TITLE: **THERMAL-HYDRAULIC SPECIFICATION** — H100 SXM5 700 W / PG25 / DESIGN POINT AND SENSITIVITIES			SCALE: **NONE** — THIRD ANGLE PROJECTION	UNITS: **mm**
DRAWING No. **CP-103**			REV **A**	SHEET **4 / 6**
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-10